STAT 709: Homework 7

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All four exercises are required. Complete every part of each exercise.

1. Modes of convergence

Jun Shao, *Mathematical Statistics*, second edition, Chapter 1:

- Exercise 116: Distinctions among modes of convergence.
- Exercise 141: The Poisson limit of binomial distributions.

2. Asymptotic distributions and estimation

Bickel and Doksum, *Mathematical Statistics: Basic Ideas and Selected Topics*, Volume I, second edition, Chapter 5, Section 5.6, *Problems for Section 5.3*:

- Exercise 25: Estimating a squared population mean.
- Exercise 26: The joint asymptotic distribution of the sample mean and variance.